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MINI REVIEW

Mini review series: Current topic in Hypertension 2024

Hypertension research 2024 update and perspectives: blood pressure management

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Abstract

This mini-review article presents a comprehensive update on recent advancements in blood pressure management, based on original research published between 2023 and 2024. As part of the "*Hypertension Research 2024 Update and Perspectives*", the review integrates insights from the JSH2024 Fukuoka Declaration, emphasizing the concept of "NEO-HYPERTENSION harmonized with society". The strategies proposed in this review aim to realize this concept by promoting lifelong blood pressure management, preemptive medicine, and leveraging digital health technologies. Key strategies include adherence to treatment guidelines, overcoming clinical inertia, and the utilization of new digital tools and novel antihypertensive agents. Additionally, the review explores the significance of multidisciplinary collaboration and non-pharmacological therapies, as well as the importance of setting individualized treatment goals tailored to patients' unique backgrounds. By aligning with the JSH's vision, this review offers a fresh perspective on contemporary clinical practices and future directions. These updates aim to contribute to improved patient outcomes and global health advancements in hypertension care.

Keywords Adherence · Inertia · Digital devices · Multidisciplinary collaboration · Non-pharmacological therapies

Introduction

The Journal of Hypertension Research emphasizes the need for strategic approaches to optimize hypertension management. Building on recent advancements, this mini-review incorporates the principles outlined in the JSH2024 Fukuoka Declaration, which advocates for "NEO-HYPERTENSION harmonized with society" [1]. This concept envisions hypertension management strategies that connect individuals and organizations through a networked approach, fostering lifelong blood pressure management, preemptive medicine for all generations, and interdisciplinary collaboration. Key areas of focus include adherence to treatment guidelines, overcoming inertia in clinical practice, and the utilization of new digital tools and novel antihypertensive agents. The importance of multidisciplinary collaboration and non-pharmacological therapies is also explored, alongside the necessity of setting appropriate treatment goals tailored to individual patient backgrounds (Fig. 1). These critical strategies aim to provide insights into the latest advancements in comprehensive blood pressure (BP) management and directly contribute to the realization of "NEO-HYPERTENSION harmonized with society". The hope is that these findings will be valuable in enhancing clinical practice and improving patient outcomes in hypertension management.

Adherence to guidelines and home blood pressure monitoring

Recent guidelines strongly advocate for BP control, emphasizing home BP (HBP) rather than office BP (OBP). However, an assessment of Asian countries reveals that guideline-based HBP monitoring (HBPM) is still not satisfactorily practiced. A survey involving 7945 physicians across 11 Asian countries highlighted challenges in adopting and effectively utilizing HBPM. Despite 95.9% of physicians recommending HBPM [2], only about half acknowledged its importance, and patient compliance was notably low. Major barriers included misunderstandings regarding HBPM and concerns about the reliability of

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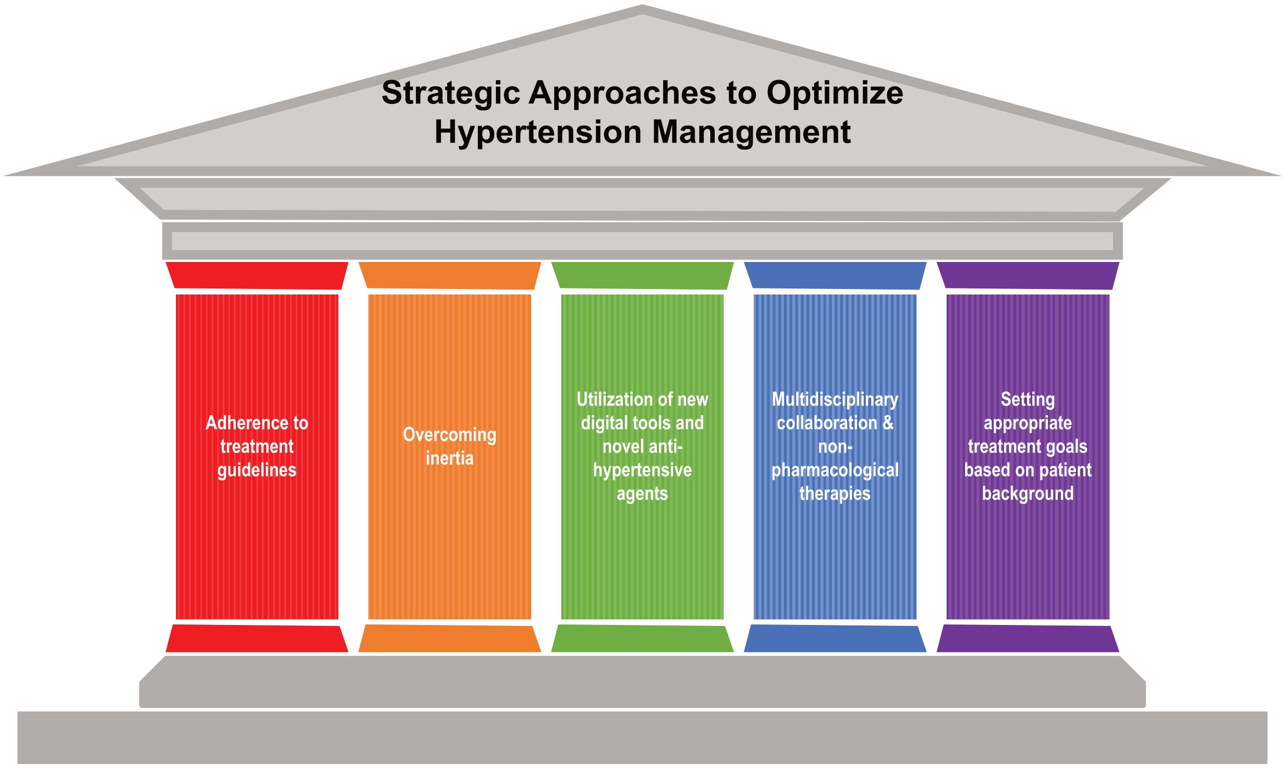
Graphical Abstract



devices. Furthermore, many physicians did not adhere to guideline-recommended diagnostic thresholds or medication timing, showcasing a significant gap between clinical guidelines and actual practice. This underscores the necessity for enhanced education and reliable tools to improve hypertension management in Asia.

Hypertension management with a new focus

Although various BP intervention methods and treatment goals for patients have been extensively discussed, there are few studies addressing the significance of residential interventions. Umishio et al. explored the impact of housing thermal environments on BP in Japan through a non-randomized controlled trial [3]. The study involved retrofitting homes with thermal insulation while measuring BP and indoor temperature. Key findings indicated that most Japanese homes are cold, indoor temperature significantly affects BP, and insulation markedly reduces BP, particularly in older adults and women. This suggests that improving living environments can effectively mitigate hypertension, comparable to lifestyle changes.

Kario et al. investigated the relationship between time in therapeutic range (TTR) for home systolic BP (SBP) and cardiovascular events among individuals with cardiovascular risk factors in the J-HOP cohort study. Over 6.3 years, participants with lower TTR (<15.3%) exhibited a significantly higher risk of cardiovascular events, including stroke, compared to those with higher TTR. A 10% decrease in TTR was associated with a 4% increase in cardiovascular events and a 9% increase in stroke risk. These findings emphasize the importance of maintaining stable, well-controlled home SBP throughout the day to reduce cardiovascular risks.

Strengthening multidisciplinary collaboration & non-pharmacological therapies

Clinicians have become too comfortable with suboptimal BP control. While Japan has a high prevalence of hypertension compared to other high-income countries, rates of recognition, treatment, and control remain low [4, 5]. Okinawa Prefecture, once renowned for its longevity, has experienced a decline due to rapid dietary changes, decreased physical activity in a car-dependent society, and rising obesity rates. A report comparing hypertension management by specialists and non-specialists in Okinawa revealed significant differences in BP control rates [6]. Patients managed by specialists achieved an average BP of

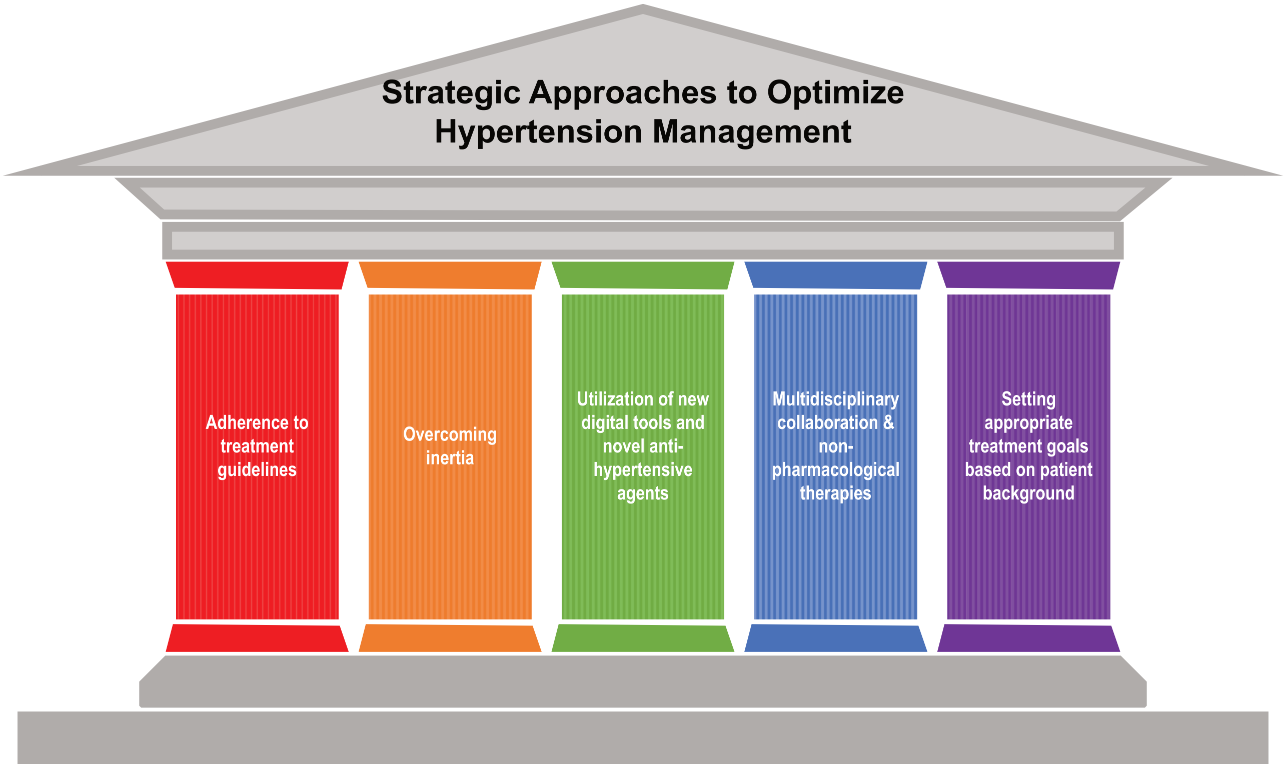


Fig. 1 Strategic approaches to optimize hypertension management

128.0 ± 15.1/73.4 ± 10.4 mmHg, with a control rate of 56.7%. In contrast, patients managed by non-specialists had an average BP of 130.1 ± 15.9/76.0 ± 10.8 mmHg and a control rate of only 46.1%. Multivariate logistic analysis demonstrated that specialist intervention and medication adherence positively influenced BP target achievement. Conversely, obesity, diabetes, chronic kidney disease (CKD), and high salt intake were identified as significant obstacles to hypertension control. To combat clinical inertia, particularly for patients with these complicating factors, the role of hypertension specialists is crucial. While hypertension management requires specialized knowledge and experience, the majority of patients are treated by general physicians. Strengthening collaboration by providing training and study sessions led by specialists is essential for enhancing the capabilities of general practitioners.

A cluster-randomized trial in China evaluated the effectiveness of a non-physician-led intensive BP intervention in reducing cardiovascular disease and mortality among hypertensive patients [7]. Conducted across 326 villages with 33,995 participants over 36 months, the intervention group demonstrated significantly greater BP reductions and a lower incidence of major cardiovascular outcomes, including myocardial infarction, stroke, heart failure, and cardiovascular-related death. The benefits of the intervention were consistent across various demographics. Although rates of hypotension were slightly higher in the intervention group, the results suggest that non-physician-led care is effective for cardiovascular risk reduction in hypertension management.

The effect of salt substitutes on BP was shown in patients with hypertension or high cardiovascular risks [8]. The DECIDE-Salt trial, a large, multicenter, cluster-randomized trial that examined the benefits of 'salt substitution' and 'salt reduction' in Chinese elderly facility residents (mean age; 71 years) with and without hypertension for 2 years. This study provided robust evidence on the effectiveness and safety of salt substitute in elderly populations, but not salt reduction. While salt substitutes compared with usual salt lowered systolic (−7.1 mmHg) and diastolic (−1.9 mmHg) BP, salt restriction had no significant effect. Cardiovascular events were reduced (HR 0.60, 95% CI 0.38–0.96) with the salt substitutes, though serum potassium levels increased slightly without adverse clinical outcomes. Mortality rates were unaffected [9]. A post hoc analysis focused on normotensive participants [10]. The hypertension incidence was significantly lower with salt substitutes compared to regular salt (11.7 vs. 24.3 per 100 person-years; HR: 0.60; 95% CI: 0.39–0.92). Importantly, salt substitutes did not increase hypotension risk (9.0 vs 9.7 per 100 person-years; RR: 1.10; 95% CI: 0.59–2.07). SBP showed a net reduction of 8.0 mmHg in the salt substitutes group. These findings suggest that merely advocating for reduced salt intake offers limited preventive benefits.

Instead, adopting salt substitutes can more effectively lower BP and reduce cardiovascular risks, particularly in older populations. Incorporating salt substitutes into public health strategies may represent a more impactful approach for managing hypertension and preventing cardiovascular diseases.

Use of wearable devices and digital tools

New medical technologies and a multidisciplinary approach to care serve as powerful tools to overcome clinical inertia. Wearable devices and smartphone applications are increasingly valuable for helping patients manage their BP and salt intake daily. Promoting self-management through digital tools can facilitate information sharing between physicians and patients, ultimately improving treatment adherence.

While YouTube has become one of the most accessible sources of health information for the general public, caution is necessary regarding the accuracy of the information presented. Kaya et al. evaluated the quality and content of 360 YouTube videos related to hypertension treatment and BP reduction, ultimately analyzing 104 videos [11]. Their findings revealed an even split, with 51% of videos deemed useful and 49% misleading. The content primarily highlighted lifestyle changes (62.5%) over pharmacological treatments (39.4%). Videos discussing alternative treatments attracted more views and likes, indicating a public preference for non-traditional therapies. This study underscores YouTube's significant role in health information dissemination, with varying reliability and the influence of video type and creator background on viewer engagement. As wearable devices become more widespread, BP management may become increasingly accessible to patients.

The HERB-DH1 trial investigated the effectiveness of the HERB Mobile App, a digital therapeutic intervention specifically for hypertension [12]. In unmedicated patients with elevated body mass index (BMI) and high salt intake, the app demonstrated a significantly greater reduction in morning home SBP compared to standard care, with an average reduction of −5.1 mmHg in patients with high BMI and salt intake. Patients who improved both their BMI and salt intake through the app experienced an even greater reduction of −7.2 mmHg [13]. Unique features of the app include daily tracking of BP, lifestyle factors, and salt intake, enabling patients to observe the direct impact of their behaviors on BP levels. This functionality empowers patients to monitor their progress, improve treatment adherence, and develop greater self-efficacy [14]. The study highlights the importance of daily engagement with the app, enhancing patients' intrinsic motivation and awareness of lifestyle factors influencing their BP. Furthermore, these apps provide educational content, daily medication reminders, and behavior modification support, contributing to improved self-management skills and increased treatment motivation [15].

Managing hypertension with wearable devices significantly enhances BP measurement accessibility for patients. Han et al. reported that a new cuffless BP device shows potential for accurate measurements [16]. They evaluated the feasibility of cuffless BP monitoring using the Samsung Galaxy Watch, involving 760 participants who contributed 35,797 BP readings over an average of 22 days. Results showed considerable variability in BP, with trends indicating higher readings on Mondays and weekday afternoons. The average calibration error for SBP was 6.8 mmHg, increasing with initial BP levels. However, many new BP measurement devices have not yet been validated for accuracy, so the use of these new technologies is not currently recommended, although they may play an important role in the future diagnosis and management of hypertension [17].

Potential for antihypertensive agents with novel mechanisms of action

Recent clinical research has highlighted the potential of Baxdrostat and Zilebesiran in addressing challenging cases of hypertension, particularly for patients with resistant or difficult-to-control hypertension. The Phase 2 BrigHTN trial evaluated Baxdrostat, a selective aldosterone synthase inhibitor, as a novel option for treatment-resistant hypertension [18]. This multicenter, placebo-controlled study involved patients already on a stable regimen of at least three antihypertensive agents. They were randomly assigned to receive Baxdrostat in doses of 0.5 mg, 1 mg, or 2 mg, or a placebo daily for 12 weeks. Results demonstrated significant dose-dependent reductions in SBP, with the highest dose (2 mg) achieving an impressive −20.3 mm Hg reduction compared to −9.4 mm Hg in the placebo group. Importantly, Baxdrostat was well tolerated, with no serious adverse events or fatalities, making it a promising candidate for resistant hypertension management. A Phase 3 trial is currently underway to further confirm its efficacy and safety profile.

Zilebesiran represents a novel RNA interference approach targeting hepatic angiotensinogen synthesis, aiming to disrupt the renin-angiotensin-aldosterone system (RAAS) at its source. In a Phase 1 study, participants received single subcutaneous doses of Zilebesiran (10–800 mg) or a placebo, with sustained follow-up over 24 weeks. Notably, doses ≥200 mg significantly lowered both systolic (>10 mm Hg) and diastolic (>5 mm Hg) BP, with effects consistent throughout the diurnal cycle and lasting up to 24 weeks [19]. This prolonged effect on BP is particularly intriguing; however, it was noted that high-salt diets reduced its efficacy, while coadministration with irbesartan enhanced BP reduction. The Phase 2 KARDIA-1 trial further confirmed Zilebesiran's efficacy in mild to moderate hypertension [20]. This trial included 394 adults randomized across various Zilebesiran dosages and placebo controls. Results indicated significant reductions in 24-hour mean ambulatory SBP at three months, with Zilebesiran showing a substantial advantage over placebo. The agent was generally well tolerated, with mostly mild adverse events, such as minor injection site reactions and mild hyperkalemia. These recent studies underscore Baxdrostat and Zilebesiran as cutting-edge advancements in hypertension treatment. By focusing on selective inhibition of aldosterone synthase or employing RNA interference to reduce angiotensinogen, these agents offer new mechanisms for sustained BP control, particularly in resistant cases. Their progress through clinical trials holds promise for future treatment paradigms aimed at challenging hypertension cases.

Appropriate antihypertensive goals depending on background

Pickersgill et al. demonstrated the impact of achieving an 80-80-80 target for hypertension management, aiming for 80% of individuals with hypertension to be diagnosed, 80% of those diagnosed to receive treatment, and 80% of treated patients to meet BP targets [21]. By 2040, widespread achievement of these goals could reduce all-cause mortality by 4–7% and prevent 110–200 million cardiovascular cases, particularly in middle-income countries, while low-income countries would see the most significant declines in disease rates. This highlights the potential of improved hypertension management to significantly reduce global cardiovascular disease burdens. Recently, large clinical studies have increasingly advocated stricter BP targets, even for the very elderly, provided physical function is maintained. In a randomized controlled trial conducted across 116 sites in China, 11,255 high cardiovascular risk participants were assigned to either intensive treatment targeting SBP less than 120 mm Hg or standard treatment targeting less than 140 mm Hg [22]. Over a median follow-up of 3.4 years, intensive treatment resulted in a lower rate of major cardiovascular events (9.7% vs. 11.1%, hazard ratio 0.88). There was no variation in effects based on diabetes status or stroke history. This study suggests that targeting a lower SBP in high-risk patients can reduce vascular events, with a minor increase in some risks. Interestingly, subgroup analysis in patients older than 70 years also indicated that strict antihypertensive treatment may be effective (hazard ratio 0.86; CI 0.69–1.07).

In a cohort study involving 36,634 VA nursing home residents, the relationship between SBP levels and cardiovascular and mortality risks was examined [23]. Results indicated that low SBP (<110 mmHg) in residents on antihypertensive therapy was associated with higher cardiovascular and mortality risks, especially in those on multiple medications. Among untreated residents, both low SBP and high SBP (≥150 mmHg) were linked to increased mortality. These findings highlight the need for tailored SBP management in nursing home residents, as aggressive BP lowering may heighten risks in this vulnerable population, underscoring a gap in evidence for optimal BP targets in long-term care. The intensity of hypertension treatment in the elderly should be reconsidered based on the patient's physical and mental functioning or the degree of care required.

Conclusion

Hypertension treatment is advancing rapidly, yet challenges remain in providing effective, patient-centered care. Traditional methods are being supplemented by flexible approaches that incorporate digital tools, such as wearable devices and home BP monitors, to enhance patient adherence and enable precise, real-time data for tailored treatment. New pharmacological developments also expand options, particularly for treatment-resistant cases. However, these innovations require infrastructure, patient education, and provider training to optimize outcomes. A balanced approach integrating evidence-based practices with new technologies and drugs can improve hypertension management, reduce cardiovascular risks, and enhance long-term patient health.

Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

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